

COVID-19 Vaccination Analysis

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Date

Abstract

Write your abstract here.

Keywords

Vaccine Efficacy, Bayesian Inference, Likelihood Inference, Likelihood Ratio Test

Introduction / Background

The rise of COVID-19 in 2019 resulted in a global crisis affecting healthcare, economics, and daily life across societies worldwide, forcing governments to implement unprecedented public health measures. By the end of 2020, countries such as the United States had seen over a million cases of COVID. This meant the race to find a reliable vaccine was on and it became a primary goal for pharmaceuticals and governments worldwide in order to control the pandemic. Pfizer and BioNTech developed BNT162b2, an mRNA based vaccine, while Janssen and AstraZeneca did not.

The clinical trial was a placebo-controlled, observer-blinded study that enrolled 43,548 participants aged 16 year or older. The participants were randomly assigned in a 1:1 ratio to receive either two doses of the BNT162b2 vaccine or a placebo, both administered 21 days apart. The primary endpoint was laboratory-confirmed COVID-19 occurring at least 7 days after the second dose (Polack et al., 2020).

The results from the trial are stated in the table below:

Group	Cases	Sample_size
BNT162b2	8	17,411
Placebo	162	17,511
Total	170	34,922

Table 1: Vaccine Efficacy Against COVID-19 at Least 7 Days After Second Dose

Statistical Methods

Model

Let X denote the number of COVID-19 cases in the vaccine group among the $n_1 = 17,411$ vaccinated with BNT162b2 participants, and let Y denote the number of cases in the placebo group

among the $n_2 = 17,511$ placebo participants. We assume

$$X \sim \text{Binom}(n_1, \pi_v) \text{ independently of } Y \sim \text{Binom}(n_2, \pi_p)$$

where π_v and π_p are the true infection probabilities in the vaccine and placebo groups, respectfully.

The parameter of interest is the vaccine efficacy:

$$\psi = 1 - \frac{\pi_v}{\pi_p} = \frac{\pi_p - \pi_v}{\pi_p}$$

Rather than working with two binomial random variables, we adopt a single binomial model. Let T denote

Likelihood Inference

Detail the likelihood approach.

Bayesian Inference

Detail the Bayesian approach.

Results

Present your findings.

Discussion / Conclusion

Discuss / conclude here.

Bibliography

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Appendix

Code

You can reference your code in the appendix (sample here).

Proofs

If applicable, include detailed mathematical derivations or additional theoretical explanations.